Research and Improvement of Skin Color Segmentation Method Based on Different Color Space

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***Abstract*—Face detection has been a more and more popular method for face recognition today. Face recognition has been used in many places such as image retrieval, video monitoring, military base and other sensitive areas of access control. There are three key steps in face recognition:face detection, feature extraction and feature matching. Face detection has been the first and the most important step in the whole process of face recognition. Skin color segmentation is an important part of face detection. Through the establishment and comparison of skin color models in three different color Spaces of RGB, HSV and YCbCr, YCbCr color space was finally determined as the ideal skin color modeling space. Some improvements are made in some way. That is to change the parameters of the transformation matrix used in the conversion from RGB space to YCbCr color space.The selected skin color region is closer to the face skin color region, so that the system can detect the face most accurately.**

Keywords-face detection; skin color segmentation

I. INTRODUCTION

In recent years, with the rapid development of science and technology, image acquisition equipment has been constantly updated, and color data sources are becoming more and more popular.A large number of scientific studies by researchers have shown that the skin tone of different parts of the body is uniform, and there is no difference in skin tone between different RACES. The reason why the skin looks different is caused by different brightness.Therefore, when in a natural color space, the face image chromaticity and brightness information separated separately, it will be a good way to achieve color segmentation.

II. COLOR SPACE

A. RGB Color Space

RGB color space is a color space established with RED, GREEN and BLUE as the coordinate system. According to the principle of three primary colors, in the RGB color space, the arbitrary color light S can be expressed as:

$$S = R(r) + G(g) + B(b) \qquad (1)$$

- Normalized RGB space

In RGB space, if the values of two pixel points $[R_1,\ G_1,\ B_1]$,$[R_2,\ G_2,\ B_2]$ are proportional, namely:

$$\frac{R_1}{R_2} = \frac{G_1}{G_2} = \frac{B_1}{B_2} \qquad (2)$$

It means that these proportional points have the same color but different brightness. Through normalization, the chromaticity component can be removed and the space of $[r, g, b]$ can be obtained, that is:

$$\begin{cases} r = \dfrac{R}{R+G+B} \\ g = \dfrac{G}{R+G+B} \\ b = \dfrac{B}{R+G+B} \end{cases} \qquad (3)$$

The normalized RGB space can better eliminate the influence of brightness change, but the detection speed of skin color is slow, which is not suitable for the occasions with high requirements on detection speed.

In the normalized RGB space, the skin color is generally concentrated in the following areas [2] :

$$\begin{cases} 0.36 \le r \le 0.465 \\ 0.28 \le g \le 0.363 \end{cases} \qquad (4)$$

B. HSV Color Space

RGB color space is hardware-oriented, and HSV color space is a kind of color space for mass users. The following is the three-dimensional coordinate representation diagram of HSV space:

Among them:

H(Hue),the value range is from 0o to 360o,start with red and go counterclockwise. The distance between R,G and B is 120o;

S(Saturation),the value range is from0.0 to 1.0;

V(Value),the value range is from0.0(black) to 1.0(white).

The three-dimensional representation of the HSV model can be imagined as that the hexagonal boundary seen by the RGB cube along its diagonals from white to black is color H, the horizontal axis is purity S, and the vertical axis is brightness V. The formula for converting RGB space to HSV space is shown below:

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$Set\ K_1 = \max\{R,G,B\}, K_2 = \min\{R,G,B\}, then$:

$$V = K_1 / 255 \tag{5}$$

$$S = \begin{cases} 0, & if(K_1 = 0) \\ 1 - \dfrac{K_2}{K_1}, & if(K_2 \neq 0) \end{cases} \tag{6}$$

$$V = \begin{cases} \left(0 + \dfrac{G-B}{K_1 - K_2}\right) \times 60, if(S \neq 0, K_1 = R) \\ \left(2 + \dfrac{B-R}{K_1 - K_2}\right) \times 60, if(S \neq 0, K_1 = G) \\ \left(4 + \dfrac{R-G}{K_1 - K_2}\right) \times 60, if(S \neq 0, K_1 = B) \\ \text{No definition}, \quad if(S=0) \end{cases} \tag{7}$$

According to the results obtained in the literature[1], The distribution range of skin color which is widely used in HSV color space is as follows:

$$\begin{cases} 0 \le \mathrm{H} \le 50 \\ 0.20 \le S \le 68 \\ 0.35 \le V \le 1.0 \end{cases} \tag{8}$$

C. YCbCr color space

YCbCr color space is also a common color space. In YCbCr color space, Y component is the brightness of pixels, CbCr is the chromaticity component, where Cb is the blue component and Cr is the red component. Therefore, brightness and chromaticity are separated from each other in this color space. YCbCr color space is converted from RGB color space, and the conversion formula is as follows:

$$\begin{bmatrix} Y \\ Cb \\ Cr \\ 1 \end{bmatrix} = \begin{bmatrix} 0.2990 & 0.5870 & 0.1140 & 0 \\ -0.1687 & -0.3313 & 0.5000 & 0.5 \\ 0.5000 & -0.4187 & -0.0813 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} R \\ G \\ B \\ 1 \end{bmatrix} \tag{9}$$

Because brightness and chromaticity are separated in YCbCr color space, skin color can form better clustering in this color space, and meanwhile, the influence of brightness change can be reduced. In addition, YCbCr color space can be obtained by simple linear transformation of RGB color space, and the conversion speed is relatively fast.

III. INTRODUCTION OF SKIN COLOR MODEL

Color space is introduced in this paper the three above, due to the different color space respectively have different skin color modeling rules, at the same time also corresponds to the different skin color model, so, we need to be in different color space respectively to attempt to establish the space where the skin color model, and comparison analysis, face detection is the most beneficial for us to select a skin color model as the final modeling space.

A. RGB color space skin color modeling

The skin color modeling rules in RGB space are shownbelow, which can be obtained from relevant data:

Under the uniform white light irradiation, the distribution area of skin color in the RGB space is shown as follows:

$$(R > 95) AND (G > 40) AND (B > 20) AND (\max\{R,G,B\} - \min\{R,G,B\} > 15)$$
$$AND(|R - G| > 15) AND (R > G) AND (R > B) \tag{10}$$

Under the condition of external flash or sunlight illumination, the distribution of skincolor in the RGB space is shown as follows:

$$(R > 220) AND (G > 210) AND (B > 170) AND (|R - G| \le 15) AND (R > B) AND$$
$$(G > B) \tag{11}$$

Due to the diversity of skin color images, the skin color modeling rules in RGB color space are as follows:

$$Rule: \text{type}(10) \cup type(11) \tag{12}$$

Where, $\cup$ stands for logic or.

As shown in the following figure, it is an example of skin color modeling in RGB color space for a face image:

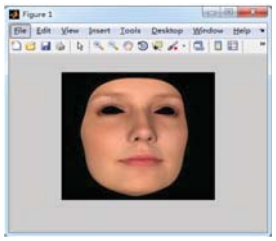

Figure 1. the original image

The image obtained after skin color modeling in RGB color space is as follows:

Figure 2. R component

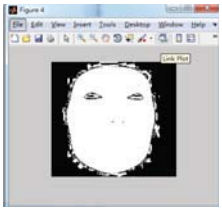

Figure 3. Gcomponent

Figure 4. Bcomponent

Figure 5.RGB Skin extract

B. HSV color space skin color modeling

When establishing the skin color model in the HSV color space, the pixel value of H's skin color point distribution was obtained by making statistics on more than 1.2 million skin color points of 100 different color face images and sampling, as shown in the figure below:

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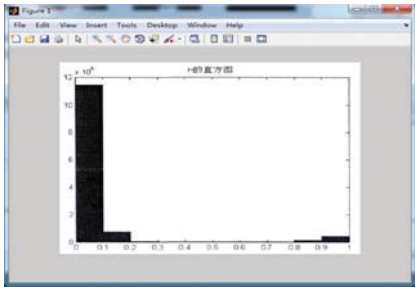


Figure 6.Probability distribution of skin tone points in hue

The tonal value is taken as the segmentation feature of skin color. After statistical analysis of skin color tone, it can be obtained that the distribution area of skin color is as follows:

$$H \le 0.2 \quad (13) \quad \text{or} \quad 0.8 \le H \le 1 \quad (14)$$

Therefore, the skin color modeling rules in HSV space are as follows:

$$Rule : \text{type}(13) \cup type(14) \qquad (15)$$

Where, $\cup$ stands for logic or.

As shown in the following figure, it is an example of skin color modeling for a face image in HSV color space:

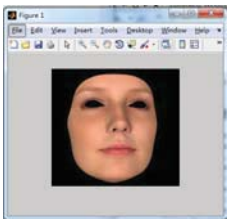


Figure 7. the original image

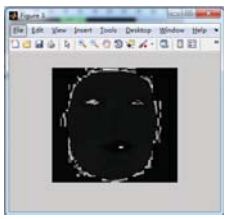


Figure 8.H component

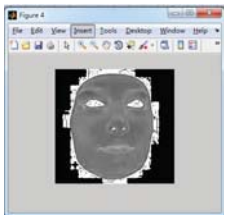


Figure 9.S component

Figure10.Vcomponent

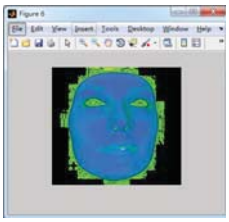


Figure11. HSV Skin extract

C. *YCbCr color space skin color extraction*

After statistical analysis of more than 1.2 million skin color pixels of 100 different color face images, the skin color curve of YCbCr is shown as follows:

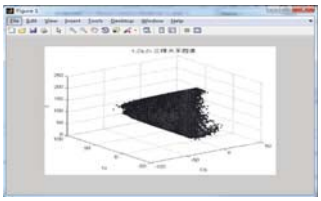


Figure 12. Three-dimensional distribution of skin color on y-cb-cr

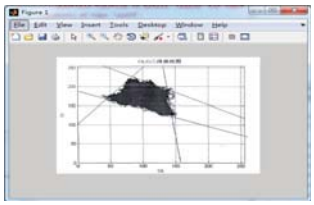


Figure 13. The two-dimensional distribution of skin color on cb-cr

By statistical analysis of the cb - cr subspace, it can be obtained that the skin color is generally distributed in the region enclosed by four straight lines :

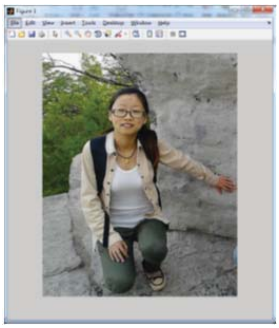
$$\begin{cases} Cr \le 1.4966 \times Cb + 96.67 & (1) \\ Cr \ge -0.3947 \times Cb + 183.33 & (2) \\ Cr \le -5.0989 \times Cb + 884.316 & (3) \\ Cr \le -0.59 \times Cb + 270.54 & (4) \end{cases} \qquad (16)$$

Therefore, the rules for skin color modeling in the YCbCr color space are as follows :

$$R : type(1) \cup type(2) \cup type(3) \cup type(4) \qquad (17)$$

Where, stands for logic or.

As shown in the following figure, it is an example of skin color modeling for a face image in the YCbCr color space:



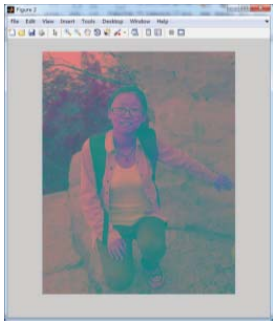


Figure 14. the original image Figure 15. RGB converted to YCbCr image

Figure 16.Y component

Figure 17.Cb component

Figure 18.Cr component

Figure 19.YCbCr Skin extract

IV. COLOR MODELING SPACE CONTRAST

These three color spaces above were used to establish different skin color models. Comparing the three skin color models, we could get some points as follows:

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1.RGB color space skin color modeling defects are more obvious;

2.The transformation from RGB color space to HSV color space is a nonlinear transformation with high computational complexity, which will reduce the detection efficiency of the detection system;

3.It is a linear transformation from RGB color space to YCbCr color space. Moreover, in YCbCr color space, brightness and chromaticity are separated, which has great advantages.

From what has been discussed above, YCbCr color space was selected as the final color modeling space in this paper.

V. IMPROVED COLOR MODELING SPACE

It can be obtained through research that in the RGB color space, the values of R, G and B can be calculated respectively, and for our face image, the size of the three values has the following relationship: R>G>B, for the three components, the greater the value of the component, the greater the influence on the face color distribution. Therefore, this paper proposes a new YCbCr color space method for detecting facial features of skin color, which can replace the original YCbCr color space, effectively improve some disadvantages of the original space, and obtain higher detection rate and lower detection rate of omission.

A. *Improved YCbCr color space*

According to the previous introduction, the formula for the conversion from RGB space to YCbCr color space is as follows:

$$\begin{pmatrix} Y \\ Cb \\ Cr \end{pmatrix} = \begin{pmatrix} 16 \\ 128 \\ 128 \end{pmatrix} + \begin{pmatrix} 65.481 & 128.553 & 24.966 \\ -37.797 & -74.203 & 112 \\ 112 & -93.786 & -18.214 \end{pmatrix} * \begin{pmatrix} R \\ G \\ B \end{pmatrix} \quad (18)$$

Where, Y represents the brightness of the image, and Cb and Cr represent the blue and red components of the image respectively.

The following three expressions can be further obtained from the above formula:

$$Y = x * R + y * B + z * G \quad (19)$$

$$Cb = \frac{1}{2(1-y)*(B-Y)} \quad (20)$$

$$Cr = \frac{1}{2(1-x)*(R-Y)} \quad (21)$$

In the above equation, x and y satisfy the following relation: $x + y + z = 1$. So we can evaluate the fourth component:

$$Cg = \frac{1}{2(1-z)*(G-Y)} \quad (22)$$

In combination with the agreement of $BT.601 in ITU-R$, This article makes $x = 0.299, y = 0.114$. And we can calculate that $z = 0.578$。

The improved new color space is shown below:

$$\begin{pmatrix} Y \\ Cg \\ Cr \end{pmatrix} = \begin{pmatrix} 16 \\ 128 \\ 128 \end{pmatrix} + \begin{pmatrix} 65.481 & 128.553 & 24.966 \\ -81.085 & 112.000 & -30.915 \\ 112.000 & -93.786 & -18.214 \end{pmatrix} * \begin{pmatrix} R \\ G \\ B \end{pmatrix} \quad (23)$$

B. *New color space skin color modeling*

In this paper, 200 face images of different ages and genders were selected, in which skin color included different body regions. Convert R, G and B of each pixel in the skin area into YCgCr, and then project these pixels into cg-cr color space to get the following figure [3] :

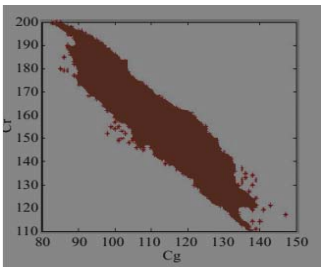

Figure 20.Cg - Cr space

As can be seen from the above figure, the brown area is the clustering area of skin color. In the image, if the pixels of Cg and Cr fall into the brown area, the system considers them to belong to the skin color area; otherwise, they do not belong to the skin color area [3]. The range of the brown area can be determined as follows, as shown below:

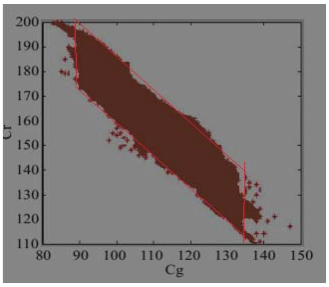

Figure 21. Cg - Cr space

As shown in the figure above, the brown region is within the region composed of four straight lines, which can be represented as follows:

$$\begin{cases} Cr = -Cg + 260 \\ Cr = -Cg + 280 \\ Cg = 85 \\ Cg = 135 \end{cases} \quad (24)$$

$$that\ is\ Cr \in [-Cg + 260, -Cg + 280]$$

$$Cg \in [85,135]$$

When the pixel value falls within the above range, it is considered to be skin color pixel; otherwise, it is non - skin color pixel.

After establishing the skin color model of the improved skin color region, it is necessary to extract the skin color region from the detection image. After the skin color is extracted, the following skin color segmentation is required.

VI. SKIN COLOR SEGMENTATION

Through the above analysis, a skin color model has been established in the improved YCbCr color space to separate skin color from background, and the method of establishing

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skin color model in the improved YCbCr color space is actually a threshold segmentation method.

The essence of the threshold segmentation method is the data of image first, to determine a threshold segmentation, the threshold, then, will the statistical information of the image to the threshold as the cut-off point, divided into two categories, and respectively to these two kinds of different values, to display the image, this is the whole process of threshold segmentation [4], in this way, the most important is to through a lot of statistical experiments, selecting a suitable threshold, can make the image more accurately the color of skin area apart from the color of skin area[5].

When skin color was segmented in the new YCbCr color space above, the selected face image background was relatively simple. When the background of the image was slightly complex, the skin color model was established and separated in the new YCbCr color space according to the above method, and the results were as follows:

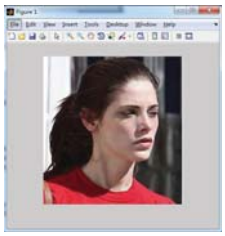

Figure 22. the original image

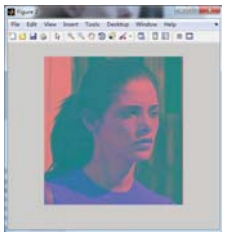

Figure 23. RGB converted to YCbCr image

Figure 24.Y component

Figure 25.Cb component

Figure 26.Cr component

Figure 27.YCbCr Skin extract

It can be seen from the above figure that no matter the face image with a single background or the face image with a complex background, after the modeling of skin color in the YCbCr color space, the skin color region and non-skin color region can be distinguished relatively well.

In the above two cases, when people's faces are not covered by glasses, beards and other things, when people's

Faces are coverd by such things,the results of skin color modeling in YCbCr color space are as follows:

Figure 28. the original image

Figure 29. RGB converted to YCbCr image

Figure 30.Y component

Figure 31.Cb component

Figure 32.Cr component

Figure 33.YCbCr Skin extract

It can be seen from the above figure that in the YCbCr color space, the shaded face image can also accurately distinguish the skin color area from the non-skin color area.

VII. SUMMARY

This paper introduces three kinds of different color spaces, and has established the skin color model respectively to make skin color extractions. Finally we chose the YCbCr color space as the target color space. For some of the deficiencies in YCbCr color space, we put forward its own improvements, that is change parameters of transformation matrix in the process of transformation. Through a series of experiments, we can find that the system can extract the facial skin color area more accurately. Thus, the detection rate of the whole system is improved.

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